signal.sci

```
clc;
clear;
// Input sequence (length should be power of 2)
x = [1, 2, 3, 4, 0, 0, 0, 0];
N = length(x);
X = fft(x, -1); // Scilab built-in FFT (DIT equivalent)
// Display results
disp("FFT (DIT) Result:");
disp(X);
subplot(2,1,1);
plot(0:N-1, abs(X), 'r');
xlabel('Frequency Index');
ylabel('Magnitude');
title('Magnitude Spectrum (DIT FFT)');
subplot(2,1,2);
phaseX = atan(imag(X), real(X)); // Phase in radians
plot(0:N-1, phaseX, 'b');
xlabel('Frequency Index');
ylabel('Phase (radians)');
title('Phase Spectrum (DIT FFT)');

```

